BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer's advice turn—an episode incompleteness failure under lexical interference. BIMS-LEGAL indexes Chinese consultation logs in a dual store and uses Soft O2 session binding with attenuated score inheritance. Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. Soft O2 improves over FlatIP on a shared turn-level index; Soft O2-C and gated Hybrid extend binding under Mix off-session gold. We report Answer Hit, Episode Completeness, graded nDCG, a failure taxonomy, and paired significance tests.

CRediT authorship contribution statement

Linrui Xu: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing. **Linrui Han:** Methodology, Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code, processed evaluation corpora, and primary result files are publicly available at `https://github.com/Kilimajaro/soft-episode-binding-legal-memory`. Upstream legal corpora remain under their original licences: CAIL2024 consultation tracks; Hugging Face `ShengbinYue/DISC-Law-SFT` and `Skepsun/lawyer_llama_data`. Code in the repository is released under the MIT Licence; redistributors of derived corpora should consult each upstream licence.

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Declaration of generative AI use in research methods

Large language models are used as generator and independent judge in the dual-protocol QA audit (N=270); prompts and settings are documented in the manuscript Methods /Discussion. No generative AI system was used to fabricate experimental numbers.